



상위 1% 월급쟁이를 위한



지상편

쿠버네티스 어나더 클래스

Application 기능을 이해하기 - Pod (probe)

쿠버네티스 첫 오브젝트 잘 끼우기

Object 그려보 이해하기

❁ Application 기능으로 이해하기

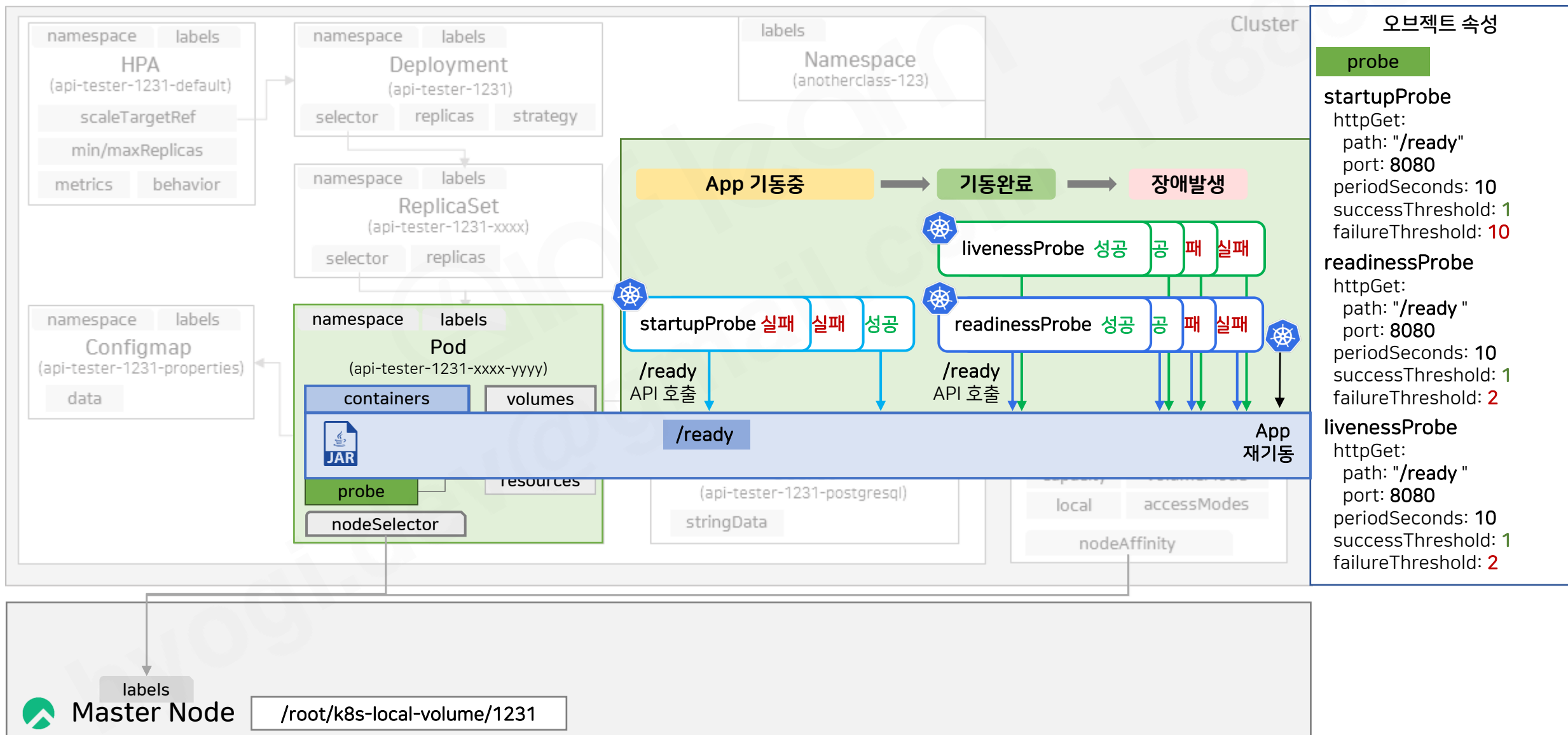
- └ Pod (probe)
- └ Configmap, Secret
- └ PV, PVC
- └ Deployment (strategy)

Component 동작으로 이해하기



Pod (probe) - 프로브 기본 개념

Application 기능으로 이해하기



Pod (probe) - Application 로그를 통한 프로브 동작 분석

Application 기능으로 이해하기

```
2023-07-28T22:48:48.: Starting service [Tomcat]
2023-07-28T22:48:48.: Starting Servlet engine: [Apache Tomcat/10.1.8]
2023-07-28T22:48:50.: Initializing Spring embedded WebApplicationContext
2023-07-28T22:48:50.: Root WebApplicationContext: initialization completed
2023-07-28T22:48:57.: Tomcat started on port(s): 8080 (http) with context
2023-07-28T22:49:04.: [System] App is initializing
2023-07-28T22:49:07.: Initializing Spring DispatcherServlet 'dispatcherServlet'
2023-07-28T22:49:07.: Initializing Servlet 'dispatcherServlet'
2023-07-28T22:49:09.: Completed initialization in 0 ms
2023-07-28T22:49:09.: [Kubernetes] startupProbe is Failed-> [System] isAppLive: false
2023-07-28T22:49:09.: [System] Database is connecting
2023-07-28T22:49:09.: [System] Database is connected
2023-07-28T22:49:12.: [Kubernetes] startupProbe is Failed-> [System] isAppLive: false
2023-07-28T22:49:12.: [System] App is starting
2023-07-28T22:49:14.: [Kubernetes] startupProbe is Failed-> [System] isAppLive: false
2023-07-28T22:49:14.: [System] App is started
2023-07-28T22:49:19.: [Kubernetes] startupProbe is Succeed-> [System] isAppLive: true
2023-07-28T22:49:19.: [Kubernetes] readinessProbe is Failed-> [System] isAppReady: false
2023-07-28T22:49:24.: [System] ConfigMap data is loading..
2023-07-28T22:49:24.: [Kubernetes] readinessProbe is Failed-> [System] isAppReady: false
2023-07-28T22:49:24.: [Kubernetes] livenessProbe is Succeed-> [System] isAppLive: true
2023-07-28T22:49:29.: [System] ConfigMap data is loading..
2023-07-28T22:49:34.: [System] ConfigMap data is loading..
2023-07-28T22:49:34.: [Kubernetes] readinessProbe is Failed-> [System] isAppReady: false
2023-07-28T22:49:34.: [Kubernetes] livenessProbe is Succeed-> [System] isAppLive: true
2023-07-28T22:49:34.: [System] Data loading is completed
2023-07-28T22:49:34.: Started AppApplication in 68.309 seconds
2023-07-28T22:49:44.: [Kubernetes] readinessProbe is Succeed-> [System] isAppReady: true
2023-07-28T22:49:44.: [Kubernetes] livenessProbe is Succeed-> [System] isAppLive: true
2023-07-28T22:49:54.: [Kubernetes] livenessProbe is Succeed-> [System] isAppLive: true
2023-07-28T22:49:54.: [Kubernetes] readinessProbe is Succeed-> [System] isAppReady: true
```

App 초기화

5초

완료

User 초기화

완료

10초

startupProbe 실패

startupProbe 실패

startupProbe 실패

startupProbe 실패

startupProbe 성공

readinessProbe 실패

livenessProbe 성공

readinessProbe 성공

livenessProbe 성공

실제 App 상황

기동되기 전엔 API를
받지 못함
- WAS(tomcat)를
사용할 경우
access.log에 기록됨

동작 확인을 위해
임의로 코드 구성

오브젝트 속성

probe

startupProbe

httpGet:
path: "/startup "
port: 8080
periodSeconds: 5
successThreshold: 1
failureThreshold: 24

readinessProbe

httpGet:
path: "/readiness "
port: 8080
periodSeconds: 10
successThreshold: 1
failureThreshold: 3

livenessProbe

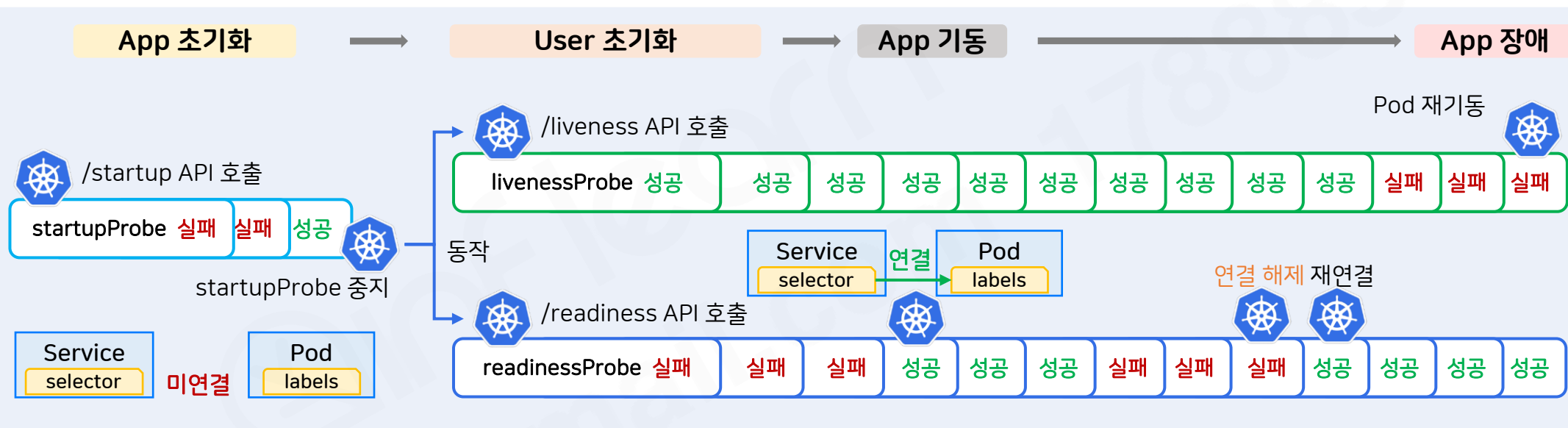
httpGet:
path: "/liveness "
port: 8080
periodSeconds: 10
successThreshold: 1
failureThreshold: 3



Pod (probe) - Application 동작 중심의 프로브 이해

Application 기능으로 이해하기

Kubernetes 제공 기능



자동화 요구사항

[API를 받을 수 없는 상태]

- App 상태 체크 (초기화가 끝났는지?)
- 외부 API 접근 **금지**

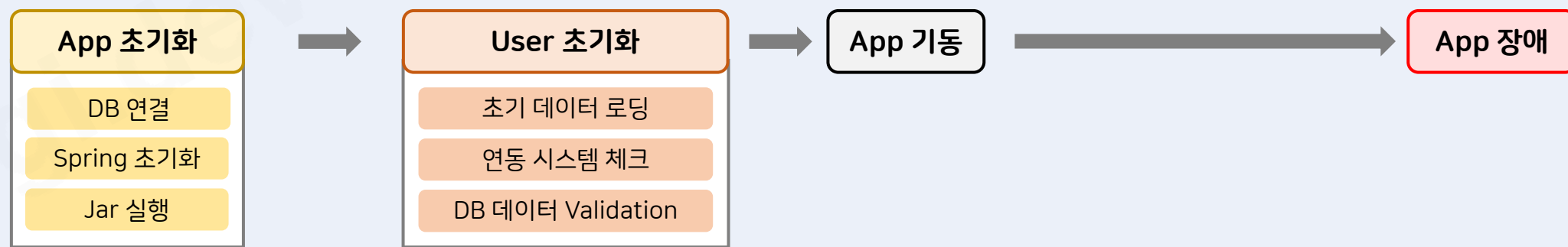
[API를 받을 수 있는 상태]

- App 상태 체크 (App이 살아있는지?)
- 외부 API 접근 **금지**

- App 상태 체크 (App이 살아있는지?)
- 외부 API 접근 **허용**

- App 재기동

Application 동작

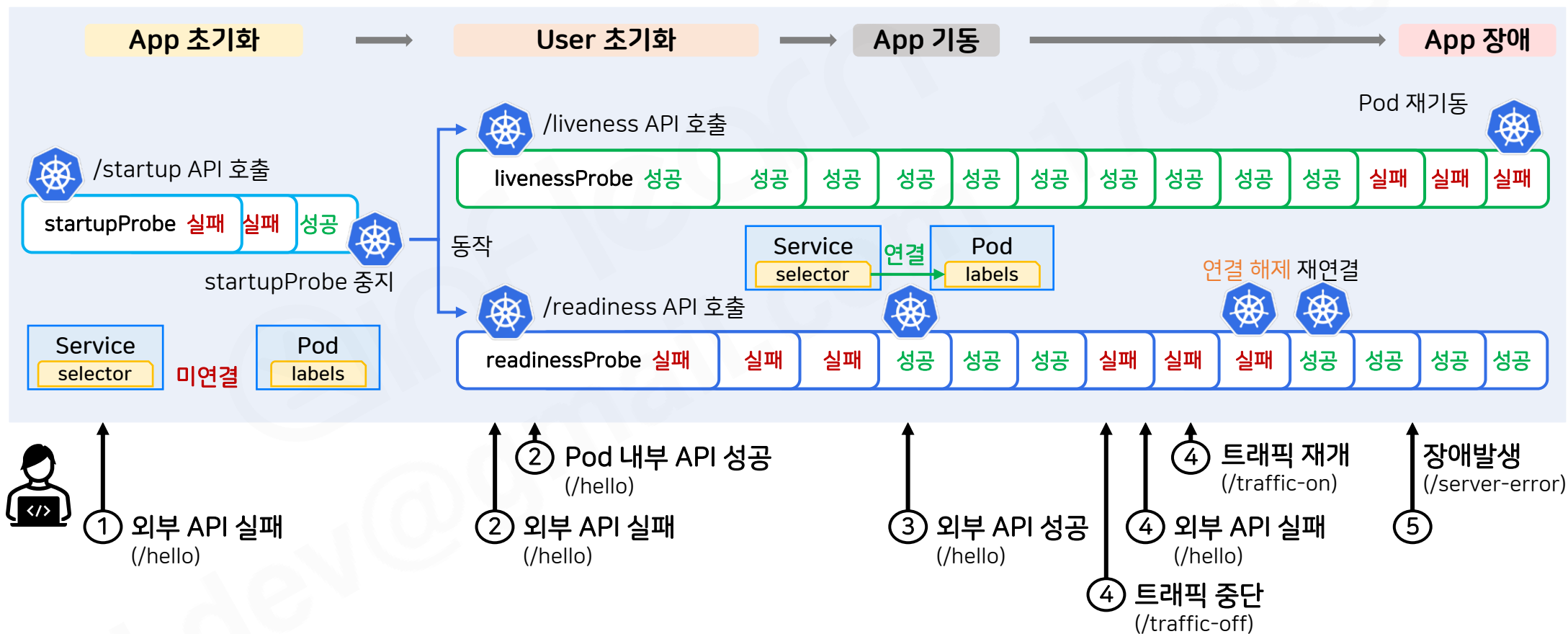




Pod (probe) - API 날려보며 프로브 동작 확인하기

Application 기능으로 이해하기

Kubernetes
제공 기능



Pod (probe) - 일시적 장애 상황에서의 프로브 활용

Application 기능으로 이해하기

